

BUSINESS REQUIREMENT DOCUMENT (BRD)

Project: Core Banking Customer 360 & Hyper-Personalization Engine

Document Version: 1.2

Status: Draft / Ready for Demo Implementation

Target Architecture: AS400 → Cloud Composer (Airflow Orchestrator) → Google Cloud Dataform (SQL Transformation & Lineage) → Google Cloud BigQuery (Medallion Architecture with Column-Level Security)

1. Executive Summary & Project Background

1.1 Project Overview

Modern financial institutions are sitting on vast treasure troves of customer data, often locked in legacy core systems like IBM AS400 (DB2). This project aims to design and implement a modern Data Warehouse (DWH) demo in Google Cloud BigQuery using a Medallion Architecture (Bronze, Silver, Gold).

By liberating transactional, credit card, and loan data from AS400, sanitizing it, and modeling it into a unified Customer 360 view, the bank will be able to run hyper-personalization workloads (such as Next-Best-Action, predictive churning, and tailored credit offers).

1.2 Strategic Objectives

- **Legacy Liberation:** Define a repeatable pattern for extracting data from legacy AS400 schemas.
- **Modernized Pipeline Orchestration:** Use Google Cloud Composer (Apache Airflow) as the centralized enterprise orchestrator.
- **Structured Data Journey (Medallion):** Implement an audit-friendly, performant, and clean three-tier storage framework.
- **Business Enablement:** Deliver a rich, analytical "Customer 360" layer ready for business intelligence (Looker/PowerBI) and machine learning (Vertex AI).

2. Business Use Case: Customer 360 & Hyper-Personalization

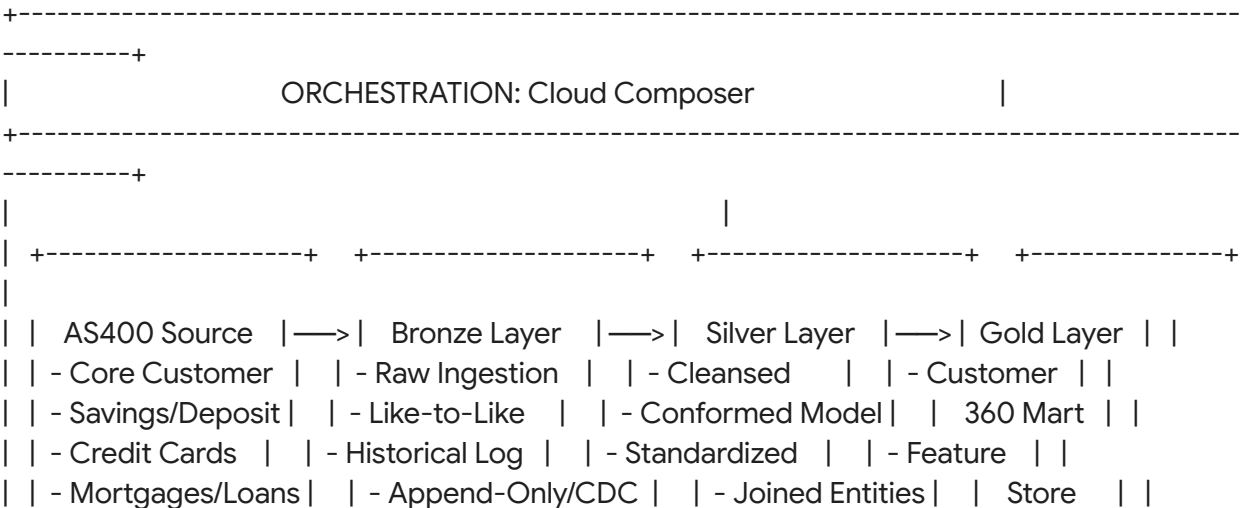
The primary objective of the target schema (Gold Layer) is to drive a **Hyper-Personalization**

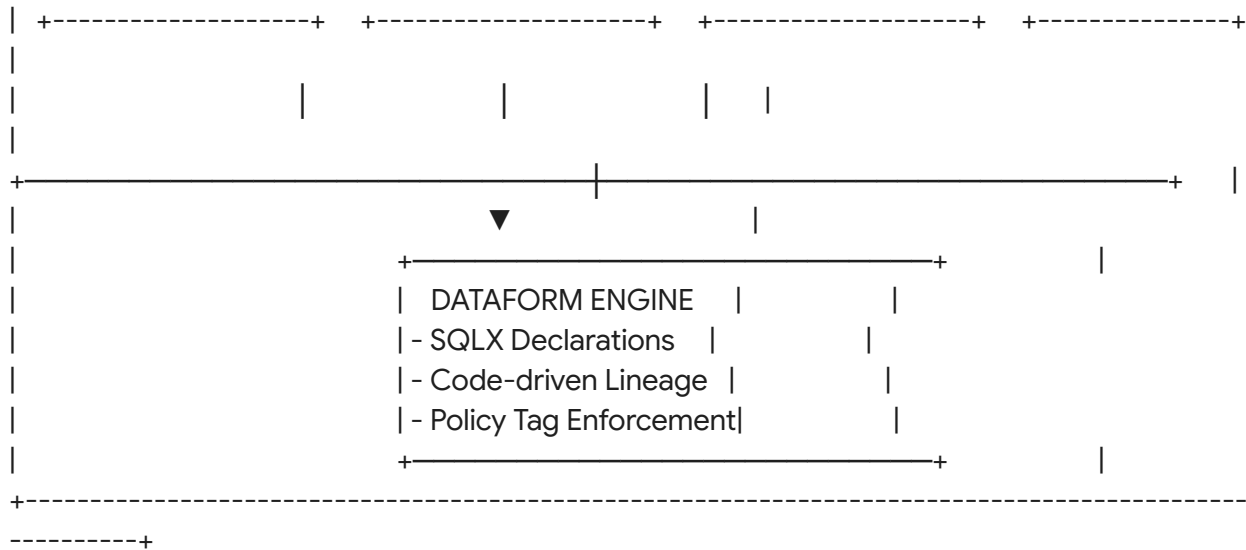
Engine. This engine supports several high-value business scenarios:

Use Case	Data Requirements	Business Value
Next-Best-Action (NBA)	Transaction patterns + Credit Card usage + Active loan status.	Recommending a high-yield savings account or a specific rewards card based on spending habits.
Credit Scoring & Pre-Approvals	Current deposits + Debt Service Ratio (DSR) + Loan repayment history.	Automating pre-approved mortgage or personal loan offers directly to mobile apps.
Churn Risk Mitigation	Sudden drop in savings balances + Dormant credit card activity.	Activating defensive retention campaigns and loyalty offers before the customer defects.
Life-Stage Marketing	Age (Birth date) + Balance changes (e.g., wedding, baby, home buying indicators).	Pitching educational savings programs, car loans, or home insurance at the right lifecycle moment.

3. High-Level Architecture Overview

The system transitions from an on-premise relational database schema (flat files / DB2 on AS400) to an enterprise-grade cloud analytical warehouse.





3.1 Architecture Components

1. **Ingestion & Stage (Pre-Bronze):** AS400 data is extracted (simulated via mock files) and uploaded to Google Cloud Storage (GCS).
2. **Orchestrator:** Cloud Composer triggers raw ingestion files from GCS to BigQuery (Bronze) and subsequently invokes the Dataform API to run compilation and executions for downstream transformations.
3. **Bronze Layer (Raw):** Direct copy of source tables inside BigQuery with minimal conversion, retaining original datatypes and adding system metadata.
4. **Silver Layer (Conformed):** Structured data model where data is deduplicated, standard date formats are applied, Null values are handled, and master-data tables are established. Managed entirely using Dataform SQLX files (type: "table" or type: "incremental").
5. **Gold Layer (Mart):** Fully aggregated, business-ready tables optimized for fast query performance and analytical applications. Dataform orchestrates these models, validating relationships and applying policy tags prior to deployment.

4. AS400 Source Data Specifications

The demo will consume 4 distinct business areas from the legacy system. The schemas below reflect common AS400 design patterns (using abbreviated column headers, non-standard date formats, and strict legacy structures).

4.1 Customer Data (AS400_CUST_MAST)

- **Description:** Legacy Customer Information File (CIF).
- **Source Format:** Fixed-width or CSV exported from DB2.

Column Name	Data Type	Sample Data	Business
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			Description
CMCIF	VARCHAR(10)	0019283746	Unique Customer Identifier (CIF)
CMNAME	VARCHAR(100)	John Doe Smith	Customer Full Name
CMADR1	VARCHAR(150)	123 Financial Way, Suite A	Address Line 1
CMDOB	NUMERIC(8,0)	19850412	Date of Birth (Format: YYYYMMDD as Number)
CMSEG	VARCHAR(5)	HNW	Customer segment (HNW = High Net Worth, MASS, etc.)
CMPHNE	VARCHAR(15)	+1-555-0199	Contact Number

4.2 Banking Accounts Data (AS400_SVDP_MAST)

- **Description:** Core Savings (SV) and Term Deposit (DP) ledger.

Column Name	Data Type	Sample Data	Business Description
ACCNO	VARCHAR(20)	SV-990088112	Account Number
AC_CIF	VARCHAR(10)	0019283746	Customer CIF Foreign Key
ACTYPE	VARCHAR(2)	SV	Account Type (SV = Savings, DP = Deposit)
ACBAL	DECIMAL(15,2)	45230.75	Current Account Balance
ACOPDT	NUMERIC(8,0)	20150920	Account Open Date

			(Format: YYYYMMDD)
ACSTAT	VARCHAR(1)	A	Status (A = Active, I = Inactive, C = Closed)

4.3 Credit Card Data (AS400_CC_TXN)

- **Description:** Log of credit card transactions and payments.

Column Name	Data Type	Sample Data	Business Description
TXNID	VARCHAR(20)	TXNCC99887766	Unique transaction ID
CRDNO	VARCHAR(16)	4111222233334444	Masked Credit Card Number
CC_CIF	VARCHAR(10)	0019283746	Customer CIF Foreign Key
TXNDT	NUMERIC(8,0)	20260615	Transaction Date (Format: YYYYMMDD)
TXNAMT	DECIMAL(12,2)	150.25	Transaction Amount
TXN_CAT	VARCHAR(15)	RETAIL	Spending Category (RETAIL, GROCERY, TRAVEL)
TXNTYP	VARCHAR(1)	D	Type (D = Debit / Purchase, C = Credit / Payment)

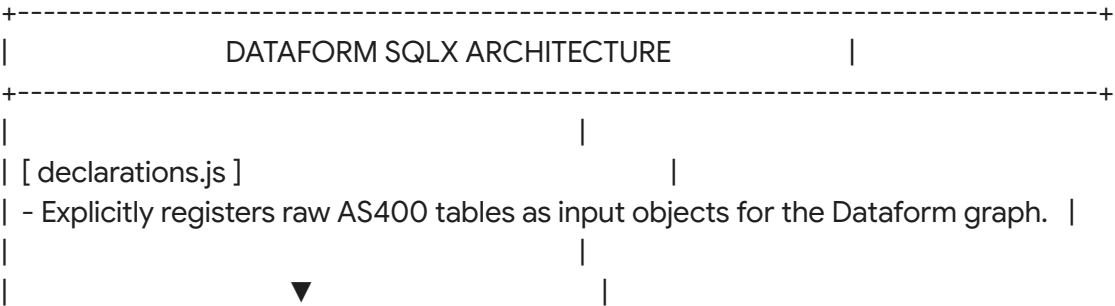
4.4 Loan Data (AS400_LOAN_MAST)

- **Description:** Loan ledgers containing mortgage and personal financing metrics.

Column Name	Data Type	Sample Data	Business Description
LN_NO	VARCHAR(20)	LN-77665544	Loan Account Number
LN_CIF	VARCHAR(10)	0019283746	Customer CIF Foreign Key
LNTYPE	VARCHAR(10)	MORTGAGE	Loan Category (MORTGAGE, CAR, PERSONAL)
LN_AMT	DECIMAL(15,2)	350000.00	Approved Principal Amount
LN_BAL	DECIMAL(15,2)	285320.10	Outstanding Loan Balance
LNMTHP	DECIMAL(10,2)	1850.00	Monthly Installment Amount
LN_DUE	NUMERIC(8,0)	20260701	Next Payment Due Date (Format: YYYYMMDD)

5. Medallion Layer Specifications & Dataform Concepts

With the inclusion of Google Cloud Dataform, physical SQL files are replaced by **SQLX** files. These files declare the configuration parameters (table type, target schema, assertions, and column-level tags) alongside standard SQL queries, linking tables automatically through the `ref()` dependency function.



```

|
| [ dim_customers.sqlx ] (Silver Table with Policy Tags)
| - config { type: "table", assertions: { ... }, columns: { full_name: { ... } } } |
| - SELECT fields FROM ${ref("AS400_CUST_MAST")}
|
|
|
|
|
| [ mart_customer_360.sqlx ] (Gold Table with Inherited Policy Tags)
| - config { type: "table" }
| - SELECT aggregate_fields FROM ${ref("dim_customers")} JOIN ...
|
|
+-----+

```

5.1 Bronze Layer: Ingestion Specs

- **Dataset Name:** demo_bronze_as400
- **Dataform Approach:** Managed via static JS/SQLX configuration declarations.
- **Declarations Pattern (definitions/sources/as400.js):**

```

declare({
  database: "gcp-project-id",
  schema: "demo_bronze_as400",
  name: "AS400_CUST_MAST",
});

```

5.2 Silver Layer: Cleansing & Conforming Rules

- **Dataset Name:** demo_silver_banking
- **Dataform Configuration:** Each entity is declared as type: "table" or type: "incremental" inside its own .sqlx file.
- **SQLX Mechanics:**
 - **Dependency Management:** Use \${ref("AS400_CUST_MAST")} to automatically determine table processing execution order.
 - **Date Normalization:** PARSE_DATE('%Y%m%d', CAST(CMDOB AS STRING)) AS birth_date
 - **Naming Standards:** Explicit transformations to convert legacy columns to clean snake_case variables.

Example Dataform SQLX: definitions/silver/dim_customers.sqlx

```

config {
  type: "table",
  schema: "demo_silver_banking",
  description: "Standardized customer demographic details from legacy CIF profiles with

```

```

security tags.",
assertions: {
  uniqueKey: ["customer_id"],
  nonNull: ["customer_id", "full_name"]
},
columns: {
  customer_id: "Unique identifier representing the core customer CIF.",
  full_name: {
    description: "Customer full legal name.",
    policyTags:
["projects/gcp-project-id/locations/global/taxonomies/812345678912/policyTags/987654321098
"]
  },
  address: {
    description: "Primary residential address.",
    policyTags:
["projects/gcp-project-id/locations/global/taxonomies/812345678912/policyTags/112233445566"
]
  },
  phone_number: {
    description: "Primary contact phone number.",
    policyTags:
["projects/gcp-project-id/locations/global/taxonomies/812345678912/policyTags/223344556677
"]
  }
}
}
}

```

```

SELECT
  TRIM(CMCIF) AS customer_id,
  TRIM(CMNAME) AS full_name,
  TRIM(CMADR1) AS address,
  PARSE_DATE('%Y%m%d', CAST(CMDOB AS STRING)) AS birth_date,
  UPPER(TRIM(CMSEG)) AS customer_segment,
  TRIM(CMPHNE) AS phone_number,
  CURRENT_TIMESTAMP() AS _ingested_at,
  "AS400_CUST_MAST" AS _source_file
FROM
  ${ref("AS400_CUST_MAST")}

```

5.3 Gold Layer: Hyper-Personalization Customer 360

- **Dataset Name:** demo_gold_analytics
- **Dataform Configuration:** Structured as a denormalized table mart_customer_360.sqlx referencing conformed Silver models.

Gold Model Profile: mart_customer_360

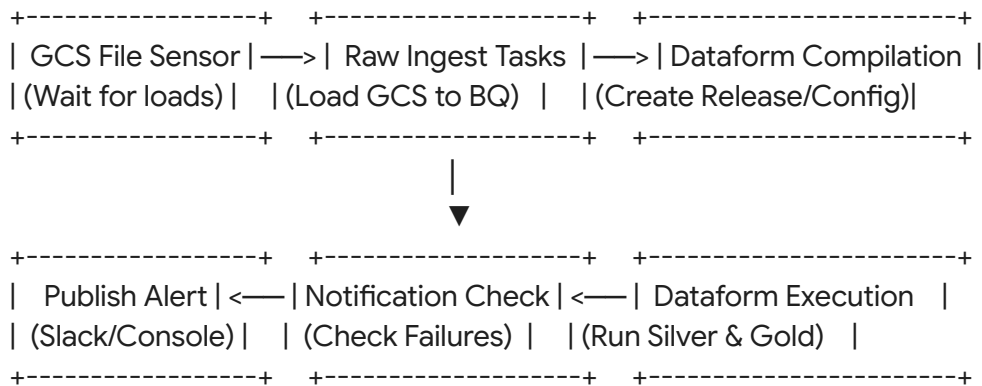
Field Name	Data Type	Derivation Logic	Hyper-personalization Purpose
customer_id	STRING	\${ref("dim_customers")}.customer_id	Primary Key
full_name	STRING	\${ref("dim_customers")}.full_name	Direct customer engagement (Inherits Policy Tag)
age	INT64	DATE_DIFF(CURRENT_DATE(), birth_date, YEAR)	Age-based segmentation
customer_segment	STRING	\${ref("dim_customers")}.segment	Tier assignment (Mass vs. HNW)
Savings Metrics			
total_savings_balance	DECIMAL	Sum of balance where account_type = 'SV'	Determines available liquidity
total_deposit_balance	DECIMAL	Sum of balance where account_type = 'DP'	Indicates investment propensity
Credit Card Metrics			
cc_spend_last_30_days	DECIMAL	Sum of txn_amount where txn_type = 'D' in last 30 days	Tracks active discretionary spending
top_spending_category	STRING	Mode of txn_category in the past 90 days	Defines next-best merchant offers

Loan Metrics			
has_active_mortgage	BOOLEAN	Check if active loan_type = 'MORTGAGE'	Screens eligibility for home refinance
total_loan_outstanding	DECIMAL	Sum of active outstanding_balance	Identifies total debt burden
Advanced Indicators			
debt_to_deposit_ratio	DECIMAL	$DDR = \frac{\text{Total Debt}}{\text{Total Deposits}}$	Risk and credit quality indicator
propensity_score_segment	STRING	Case Statement based on age, balances, and spend category	Custom demographic target bucket

6. Orchestration Workflows (Composer/Airflow & Dataform Integration)

Instead of scheduling sequential SQL scripts, Cloud Composer manages high-level task ordering and executes the entire Dataform workflow using Google Cloud integrated Airflow Operators.

6.1 DAG Workflow: daily_as400_medallion_load



6.2 Airflow DAG Operational Parameters

- **Scheduler Interval:** 0 2 * * * (Runs daily at 2:00 AM UTC).
- **Airflow Tasks with Dataform Operators:**
 - **Task 1: Compile Dataform Repo:** Uses DataformCreateCompilationResultOperator to fetch SQL code from Git, compile the dependency tree, and prepare the runtime plan.
 - **Task 2: Run Transformations:** Uses DataformCreateWorkflowInvocationOperator to execute all SQLX models from Silver to Gold. Dataform automatically processes tasks in parallel where no dependency exists.

7. Data Quality, Governance & Dataform Assertions

Because this pipeline processes Personally Identifiable Information (PII) and core transactional parameters, compliance guidelines and data quality must be strictly enforced.

7.1 Built-in Dataform Assertions

Dataform automatically creates dependency-aware verification views. If an assertion fails, Dataform sends a pipeline alert to the monitoring console, preventing invalid data from hitting the Gold layers.

- **Example Custom Assertion (definitions/assertions/assert_balanced_accounts.sqlx):**
config { type: "assertion" }
SELECT * FROM \${ref("dim_accounts")} WHERE balance < 0 AND account_type = 'SV'

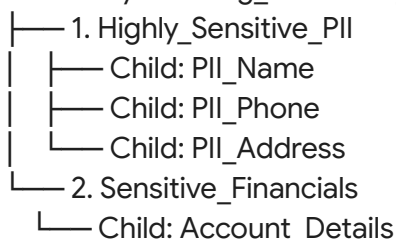
7.2 Data Governance & BigQuery Policy Tags (PII Protection)

To comply with global regulatory rules (e.g., GDPR, CCPA, PCI-DSS) and local banking guidelines, BigQuery's native **Column-Level Security (CLS)** and **Dynamic Data Masking** are applied across Silver and Gold datasets.

7.2.1 Security Taxonomy Design

We define a hierarchical, centralized Taxonomy called Banking_Customer_Data_Classification inside the Google Cloud Policy Tag Manager.

Taxonomy: Banking_Customer_Data_Classification



The specific classification mappings, enforcement locations, and data masking strategies

applied to this taxonomy are defined as follows:

Policy Tag Name	Associated Fields	Classification Level	Data Masking Strategy	Allowed Groups
PII_Name	full_name	Highly Sensitive PII	SHA-256 Hashing: Non-authorized users see a 64-character hash value.	Vetted-Market ers, Compliance-A uditors
PII_Phone	phone_number	Highly Sensitive PII	Custom Mask (Last 4 digits visible): Shows XXXX-XXXX-1234.	Vetted-Market ers, Customer-Sup port
PII_Address	address	Highly Sensitive PII	Nullify / Full Masking: Returns an empty string "" or NULL.	Compliance-A uditors
Account_Details	crdno (Credit Card)	Sensitive Financials	Regex Mask (First 12 digits masked): Shows XXXXXXXXXXXX X4444.	Risk-Managem ent

7.2.2 Cloud IAM Role Assignments & Access Control

Access to unmasked column data is strictly governed using the standard **Fine-Grained Reader** role.

- **Default Restricted Access:** Any user or service account query executing against Silver or Gold tables who is not mapped to an allowed group will automatically trigger BigQuery's data masking policy, resolving the protected columns using the strategies defined above.
- **Authorized Access:** Members of authorized Google Workspace / Cloud Identity groups must be explicitly assigned the roles/datacatalog.categoryFineGrainedReader role directly on the specific Policy Tag.

7.2.3 Dataform Integration Framework

To ensure continuous compliance, Dataform manages the application of Policy Tags declaratively. When Dataform runs compilation and executes deployment statements (DDL/DML):

1. BigQuery automatically verifies that the Service Account executing the Dataform workflow has the roles/datacatalog.categoryFineGrainedReader permission (required to write and preserve tags on downstream tables).
2. Dataform applies the policy tags directly to the schemas of the target tables during deployment, preserving security tags across table overwrites without manual database administrator intervention.

8. Mock Data Generation & Validation Plan

To execute a successful demo, we require a mock data generator that produces realistic, relational profiles across multiple systems.

8.1 Mock Customer Persona Archetypes

For demonstration purposes, the pipeline should ingest and surface these representative customer profiles:

1. **The "HNW Saver":**
 - High savings balance ($> \$250,000$).
 - No active loans.
 - Low CC transaction activity, mostly travel-focused categories.
 - *Expected Gold Output:* High Net Worth segment classification, high investment propensity recommendation.
2. **The "Squeezed Mortgage Holder":**
 - High loan outstanding (Mortgage) with substantial monthly installment payment.
 - Low-to-medium savings balance ($\$2,000 - \$5,000$).
 - High credit card utilization on grocery and utility categories.
 - *Expected Gold Output:* High debt service indicator, premium credit restructuring next-best-action.
3. **The "Active Millennial Shopper":**
 - High CC transactions across digital, retail, and restaurant categories.
 - Moderate savings balance.
 - No loan profile.
 - *Expected Gold Output:* Retail affinity tag, cash-back reward credit card recommendations.

9. Next Steps and Architecture Deployment

To validate this design in a live workspace environment, the following assets should be

prepared:

1. **Policy Tag Taxonomy Setup:** Provision the taxonomy resource via Terraform or Google Cloud Console and generate the required Policy Tag URNs.
2. **Dataform Repository Initialization:** Connect Google Cloud Dataform to a Git repository containing the SQLX project layout (dataform.json, package.json, and directories).
3. **DDL & Source Configurations:** Establish the database declarations matching legacy DB2 schemas.
4. **Mock Data Generators:** A Python routine to export simulated AS400 tables as CSV files to Cloud Storage.
5. **Airflow DAG scripts:** Production-ready DAG file calling Dataform API commands to compile and invoke pipeline runs sequentially.
6. **Customer 360 Query Views:** Analytics queries showing how personalized messaging engines can leverage the Gold tables instantly.